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Implementation of AI in career counselling for university students: a systematic review

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Artificial intelligence (AI) is transforming university career services, shifting from traditional predictive algorithms to interactive generative agents. This transition raises critical questions about student agency and the digital platformisation of career guidance. In this systematic review, we examined empirical evaluations and technical architectures of AI implementations in higher education career counselling from 2015 to 2025. Following PRISMA 2020 guidelines, we identified 43 studies across Web of Science, Scopus, and ERIC. Methodologically, we appraised study quality using the Mixed Methods Appraisal Tool (MMAT) and conducted a thematic narrative synthesis. We identified three main AI implementation typologies: (1) Generative Chatbots offering scalable 24/7 support, though limited by user trust barriers and a lack of perceived empathy; (2) Predictive Analytics forecasting employability, which require transparent, explainable models; and (3) Recommender Ecosystems that match students to personalized career pathways using institutional data. Interpreting these empirical findings through Career Construction Theory and Platformisation, we highlight a tension between technological efficiency and student agency. While AI expands information access, over-reliance on opaque algorithms introduces the risk of "automation bias," where students passively accept recommendations rather than actively constructing their vocational identities. We conclude that AI should augment, rather than replace, human counsellors within a hybrid support model. Future research should prioritize longitudinal evaluations of AI interventions, algorithmic audits of recommender platforms, and the development of Explainable AI (XAI) to foster critical AI literacy.

KEYWORDS
AI, career counselling, higher education, platformisation, student agency, systematic review

1 Introduction

Career counselling services are growing more digital in universities while remaining available in person. Career centers face demand from growing student bodies and employers requesting rapidly evolving skills. These pressures incentivize organizations to implement AI to augment career guidance practices, because AI can scale availability of information and personalization of advice. Sampson and Osborn (2014) review how career interventions integrate information and communication technology. Westman et al. (2021) summarize existing demands and future opportunities for AI applications in career guidance. Studies also highlight

implementation constraints such as siloed data infrastructures at universities, unequal access to digital technology among students, and ambiguity around professional competencies in online environments. Khurumova and Pinto (2023) research student readiness for online career guidance and computer assisted career guidance.

Studies of computer assisted career guidance systems (CACG systems) initially focused on information delivery and structured assessment. These technologies automated person environment fit by administering inventories, using occupational databases, and applying rule-based matching algorithms. Sampson and Osborn (2014) position these technologies among other career interventions throughout history that have been supported by technology. Leung (2022) discusses recent trajectories for computer assisted career guidance systems research and practice in relation to Career Construction Theory. Later CACG systems use machine learning and recommender systems to relate students' records and learning analytics data to labor market information. Herath et al. (2024) review technology and examples that support student career path planning. José-García et al. (2022) demonstrate one example of a system that recommends job roles based on student skillsets and visualizes skill gaps students must address. Modern implementations can assess skill gaps, predict employment outcomes, and intervene with learning opportunities. Qualities of interventions raise ethical considerations that are especially salient to counselling, such as limits to explainability and student privacy issues around educational data mining. Gedrimiene et al. (2023) survey learners' perspectives on AI features in learning analytics platforms to inform how they support career decisions. Guleria and Sood (2022) describe a method for creating explainable models to build trust and interpretability in educational data mining applications adapted for career counselling.

Generative conversational agents represent another development in this progression. Large language models enable open-ended question and answer dialog that approximates the exchange found in coaching and counselling conversations. This interaction marks a departure from structured assessment toward eliciting career stories. Bommasani et al. (2021) provide an overview of foundation models, which underlie many modern language technologies. Bender et al. (2021) provide a critique of foundation models focused on risks that arise when scaling language technologies to large models, including biases, opacity, and fluency that can mislead users. Concerns about reliability and trust in career guidance take on particular urgency given that students may use AI recommendations to inform their career choices. Westman et al. (2021) outline ethical challenges of trust, accountability, and responsibility when AI is used to supplement professional career guidance.

In this systematic review, we present the empirical literature on AI for university career counselling published between 2015 and 2025. This review addresses three research questions. First, what types of AI systems have been empirically implemented in career counselling for university students? Second, what pedagogical outcomes and counselling processes are reported in empirical evaluations of these systems? Third, what methodological limitations and research gaps characterize the current evidence base?

The present review provides a resource to students, administrators, practitioners, and policy-makers who wish to understand or implement AI in university career guidance. The implications of this review are practical as well as theoretical, in the sense that it provides a conceptual and evidence-based framework for the responsible integration of next-generation AI tools in higher education career services.

To improve readability and guide our multidisciplinary readership, the remainder of this manuscript is organized as follows. Section 2, Literature Review, situates the study within existing scholarship, thematically synthesizes prior work, and articulates the specific research gap. Section 3, Methodology, details our systematic review protocol, review type justification, and our approach to quality appraisal utilizing the Mixed Methods Appraisal Tool (MMAT). Section 4, Results, presents our empirical findings grouped into three identified implementation typologies. Section 5, Discussion, interprets these findings through our theoretical lenses, evaluates their contextual significance, and addresses how methodological constraints inform our synthesis. Finally, Section 6 outlines concrete Future Directions for research, followed by the study's Limitations and Conclusion.

2 Literature review

Classical CACG systems have been built around the use of machine learning classifiers and recommender algorithms, which infer fit from historical patterns of educational trajectories and occupation matches (Herath et al., 2024; Leung, 2022; Sampson and Osborn, 2014). In the contemporary literature, these systems are often described as diagnosis-driven tools that help students navigate complex data by providing instant, data-driven feedback (Herath et al., 2024; José-García et al., 2022). By default, these systems also inherit the privacy and ethical implications of traditional data systems in higher education (Herath et al., 2024). However, predictive models have been the subject of sustained empirical scrutiny. For instance, early literature describes these applications as providing a simple augmentation for information retrieval, rather than provoking deeper or more systemic learning (Sampson and Osborn, 2014). In this sense, there is a growing urgency to study AI in the context of university career counselling. Rapid AI adoption has outpaced scholarship on its educational and ethical implications (Westman et al., 2021).

Today, the field is undergoing a significant technological transition from predictive analytics to Generative AI. Generative agents, as opposed to classification models, do not simply predict "fit" but actively engage students in exploratory, co-constructive dialog (José-García et al., 2022). Instead, they offer the potential for co-constructive, bidirectional conversations, as represented by Large Language Models (LLMs). Recent studies show that generative agents can be prompted to simulate human coaches by developing custom narratives and adaptively scaffolding student reflection (Flink et al., 2025). This trend represents a departure from models that provide access to information toward models that also help students construct narrative identity. Moreover, the act of counselling is being transferred from a human-reserved task to a joint student-algorithm exercise. Bankins et al. (2024) define a parallel maturation of AI in students' engagement with career exploration and management across the career lifespan, again highlighting the central questions regarding AI and sustainable, equitable pathways through an automated labor market.

Importantly, this transition raises key questions about the balance of agency between human students and AI "partners." As vocational adaptability is a crucial predictor of career construction success (Savickas, 2013), generative AI should theoretically allow students to

better attune to labor market signals and scaffold the reflection process by automating information retrieval. In contrast, a heuristic like "automation bias" can emerge, whereby students become over-reliant on the suggestions of autonomous systems and do less of the "conceptual work" necessary to construct vocational identity (Westman et al., 2021). In this view, AI is a double-edged sword (Bankins et al., 2024) with a tradeoff between empowerment and dependency (Westman et al., 2021). There is a serious risk that high-quality interventions will not reach the student population due to structural and dispositional barriers (José-García et al., 2022).

In turn, this discussion is situated within a meta-contextual trend known as "platformisation" (Staunton, 2025). Platformisation is a global trend toward large-scale digital ecosystems that integrate and coordinate activities, which were previously performed by human development institutions. Staunton (2025) argues that platforms are socio-technical gatekeepers in this way. The design and algorithmic logics of such systems privilege certain metrics of success over others. By extension, platform-based career tools may limit student development by circumscribing exploration to opaque recommendation logic. AI is thus a multi-level, political problem that involves ethical tradeoffs and risk-averse governance. This shift also raises risks of automation at scale, whereby AI systems are deployed to discipline student populations using nudges and complex algorithms (Muhammad, 2023). In this way, AI career tools nudge students toward historically well-trodden paths.

The current literature has not yet coalesced into a unified view on these problems. Herath et al.'s (2024) recent systematic review focused on the efficacy of CACG tools, while José-García et al.'s (2022) were focused on practical considerations in system design and deployment. Likewise, earlier reviews (Pandya and Wang, 2024) have shown that the technology itself is just one piece of a broad ecosystem of organizational barriers to system implementation. As noted by the Delphi expert panel from Westman et al. (2021), empirical research has failed to keep up with rapid AI deployment.

This leaves a critical gap in the understanding of how "AI as a partner" reconfigures the core activity of career construction. While existing reviews effectively map the technical landscape, a high-quality systematic review is required to synthesize the implementation of AI in university career counselling from 2015 to 2025 through the theoretical lenses of Career Construction Theory and Platformisation.

Building on previous reviews by thematically organizing prior works around three themes: the predictive-to-generative turn in AI, issues around student agency, and platformisation as an organizing logic, this paper contributes to the field by addressing current research gaps and critically positioning itself within existing scholarship. Instead of merely describing how various tools work and evaluating their effectiveness, this review discusses how algorithmic designs reshape the ethics and mechanics of career-building.

3 Methodology

3.1 Research design

We conducted a systematic literature review in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Page et al., 2021). The review protocol, including the objectives, search strategy, inclusion criteria, and analysis plan, was pre-specified in advance to ensure transparency and reproducibility of our methods. We also used the Covidence platform to manage the screening and selection process. This provided a structured workflow for de-duplication of references, blind screening of titles/abstracts by multiple reviewers, and resolution of conflicts in study inclusion.

We selected a systematic review approach because we wanted to critically appraise and synthesize the scattered empirical evidence across various fields (Newman and Gough, 2020; Zawacki-Richter et al., 2020). Since our research aim is to identify how various AI tools are actually being used and implemented educationally, conducting a systematic review allows us to be transparent in our methods, avoid bias in selection (Gough et al., 2017), and adhere to reporting guidelines for reviews on emerging technologies in education.

3.2 Data collection

A comprehensive search of academic databases was performed for literature relevant to the research question. The systematic search was conducted between December 5 and December 25, 2025, by two independent researchers to ensure comprehensive coverage and minimize selection error. We constructed searches so they can be reproduced according to each database's unique syntax and indexing tendencies. We selected Web of Science, Scopus, and ERIC (Education Resources Information Center) as the primary data sources for our literature search, as they encompass a wide range of high-quality journal articles, conference proceedings, and educational research reports. See Table 1 for the final Boolean search strings used to query each database. We applied no language filter in the search stage (only in Covidence's screening stage), but our

TABLE 1 Exact Boolean search strings.

Database	Search query string
Scopus	TITLE-ABS-KEY (("artificial intelligence" OR "AI" OR "machine learning" OR "expert system" OR "chatbot") AND ("career counselling" OR "career guidance" OR "career advising" OR "career development") AND ("university" OR "higher education" OR "undergraduate" OR "college" OR "students" OR "youth"))
Web of Science	TS = ("artificial intelligence" OR "AI" OR "machine learning" OR "expert system" OR "chatbot") AND TS = ("career counselling" OR "career guidance" OR "career advising" OR "career development") AND TS = ("university" OR "higher education" OR "undergraduate" OR "college" OR "students" OR "youth")
ERIC	("artificial intelligence" OR "AI" OR "machine learning") AND ("career counselling" OR "career guidance") AND ("higher education" OR "college students")

inclusion criteria in practice constrained the sample to English-language publications, since these were the ones that dominated the databases.

The database search yielded 121 records from Web of Science, 189 records from Scopus, and 45 records from ERIC. To extend database searching, forward snowball searching was conducted for key authors in the field. We selected authors to follow by (1) identifying those that appear in at least two records from our initial database search and (2) selecting the authors of recently published systematic reviews addressing the research question. Publication profiles for these researchers were searched for additional work which had not yet been indexed and reference lists of these review articles were screened. This snowball searching strategy identified 10 additional records which we added for screening. All the references were then imported into Covidence for deduplication.

3.3 Screening and selection

After deduplication, 287 unique records remained. We applied relatively broad inclusion at this stage in order to err on the side of sensitivity—any study that mentioned both career counselling/guidance and some form of artificial intelligence/intelligent system, in an educational setting, was flagged for full-text review. Covidence helped manage this process by allowing the two reviewers to vote "Include" or "Exclude" on each citation, with conflicting votes requiring either discussion or a third reviewer for adjudication. The most common reasons for excluding a study at this stage were: it did not actually use an AI or other computational implementation (i.e., it was purely theoretical or a general discussion of career issues), or it involved career guidance for a population that was not in the scope of our review (e.g., only working adults, or K-12 students with no application to higher education). The full texts of all studies that passed the title/abstract screening stage were then obtained. Each full text was subsequently assessed against the inclusion criteria. For transparency and reproducibility purposes, the eligibility criteria imposed at screening were operationalized into PICOS format. The inclusion/exclusion criteria are listed in Table 2. Ultimately, 43 studies were selected (Supplementary Appendix A) for inclusion in the review for data extraction and synthesis. Figure 1 shows the summary of this process in PRISMA format (Haddaway et al., 2022).

A data extraction form was created in which key features of the included studies were recorded. These included: bibliographic information, background/context and sample, AI technology/system used, purpose/function of the system, design and methods of study, outcome measures and main results, and any author discussion of limitations or implications. One reviewer extracted data for each study, which a second reviewer cross-checked. Discrepancies were discussed to reach consensus.

Results of the methodological quality appraisal, performed utilizing the Mixed Methods Appraisal Tool (MMAT), are summarized in Table 3. Consistent with MMAT guidelines, the summary outlines the particular methodological strengths and limitations of the relevant study design categories rather than relying solely on overall numerical scores (Hong et al., 2018). From here, the field needs to go beyond self-reported metrics. The rationale for selecting MMAT was that it has been validated for appraising different study designs (qualitative, quantitative descriptive, and mixed methods) simultaneously, and educational technology research is known for being quite interdisciplinary.

TABLE 2 Inclusion and exclusion criteria.

Criterion	Inclusion criteria	Exclusion criteria
Population	Undergraduate or graduate university students; recent graduates; prospective students in a higher education context	K-12 students (primary/secondary education); working adults unrelated to university placement; general population
Intervention	Digital systems utilizing AI, Machine Learning (ML), NLP, or recommender algorithms specifically for career guidance/counselling	Digital tools without "intelligent" features (e.g., static information websites, digitized PDF tests); general AI tools not specific to career guidance
Outcome	Empirical evaluation of user outcomes or technical documentation of novel system architectures implemented in university settings	Opinion pieces, editorials, or purely conceptual proposals without any empirical implementation or data
Study type	Peer-reviewed journal articles, conference proceedings, and scholarly reports	Unpublished theses, dissertations, white papers, and non-peer-reviewed articles
Language	English	Non-English publications
Timeline	January 2015–December 2025	Published before 2015

More importantly, this MMAT appraisal informed our perspective on the review instead of merely reporting the descriptive findings. One of the major limitations in the current literature, as depicted in Table 3, is the overreliance on cross-sectional designs, and lack of longitudinal control groups across the studies. As such, our summarization of the evidence does not make direct causal claims about the pedagogical implications of AI for longer-term success of the students' career. Instead, these quality, robustness, and validity considerations limited our narrative synthesis to assessing short-term implementation typologies, technical system feasibility, and immediate user acceptance. In addition, the lack of longitudinal robustness detected in the appraisal informed our critical interpretive framework, particularly our focus on the risks of automation bias and narrative truncation in the discussion of this analysis to follow.

3.4 Evidence synthesis logic

Since the studies included in this critique were highly heterogeneous in the designs, populations, and algorithmic architectures, they were not amenable to formal statistical meta-analysis. We instead conducted a thematic narrative synthesis. In order to bring a more unified synthesis of evidence across these varying methodological traditions, we supplemented quantitative system metrics (algorithmic

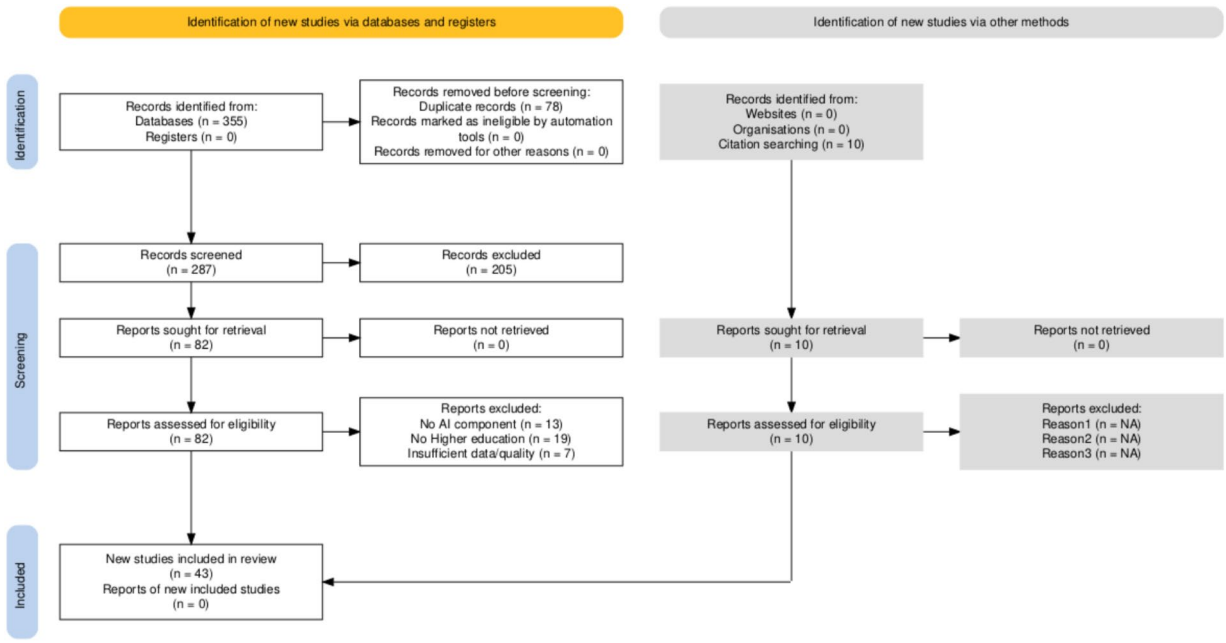


FIGURE 1
PRISMA 2020 flow diagram for study selection (generated using the PRISMA 2020 tool by Haddaway et al., 2022).

classification accuracy, user engagement rates) retrieved from technical literature with qualitative insights (perceived user trust, empathy, and agency) generated from pedagogical evaluations. These narrative integrations were mainly driven in this area by the methodological quality of the studies. Technologically descriptive studies (often lacking control groups) were used mainly to map out the technical architecture and feasibility of AI systems while higher-quality mixed-methods and qualitative studies guided our interpretive pedagogical synthesis of user behavior. Inductively, three broad thematic clusters of AI applications were identified. For each theme, we synthesize empirical evidence and discuss the broader theoretical implications.

4 Results

4.1 Theme 1: AI chatbots for career counselling

A recurrent thread was the deployment of AI-based chatbots and conversational agents to emulate human counsellors and provide students with career guidance. Of the 43 studies, over a dozen focused on chatbot-based career counselling tools, or were in part based on such systems. These systems would typically use natural language processing (NLP) techniques to allow users to converse about career topics, ask questions, and be directed to resources (sometimes including self-guided career assessments or exercises). The attraction of chatbots is their immediacy and scalability: they could be available 24/7 with real-time responses, solving issues of counsellor shortages and appointment backlogs (Akshansh and Bhavani, 2019; Suresh et al., 2021). However, chatbots are more than generic off-the-shelf designs: one recent study among others demonstrated the benefits of theory informed chatbot architectures, where a Self-Determination Theory based career counselling chatbot showed better alleviation of career decision making difficulties compared to a non-theoretical chatbot in an experimental study with university students (Han et al., 2025). Several studies deployed chatbots as a first-line support mechanism. For instance, a chatbot can be deployed as part of a university's career guidance web application or social media presence, to field common student questions and guide them to the next step (e.g., about job searching or CV tips, or wanting to know about different career options). Suresh et al. (2021) designed and developed a career counselling chatbot on Facebook Messenger using AI, which allowed students to interact with the guidance system through an interface they were familiar with; their findings suggested that such a chatbot could indeed provide effective basic guidance and free up human counsellors' time for higher-level tasks (the chatbot could respond to FAQs and direct students to relevant resources). In another study, Westman et al. (2021) conceptualized various "modes of agency" for AI in guidance, one of which is an AI agent directly advising students in a conversational style—in other words the chatbot model – and investigated student and staff perceptions of this mode.

The majority of chatbot systems in the reviewed literature were rule-based or hybrid in terms of AI design, combining scripted dialog flows for the core conversation structure but using some degree of AI to interpret user inputs. Simpler chatbots relied on keyword matching and pre-defined responses to answer FAQs, while more sophisticated systems used machine learning-based NLP to parse open-ended student questions.

Recent developments in the educational technology focus on creating intelligent chatbots that utilize AI and NLP algorithms to produce an automated, human-like conversation (Monika and Balaji, 2025). In contrast to the previously described static approach, this method enables the AI-powered tools to process unstructured students' requests, meaning that the program can more efficiently infer the user's intent and, subsequently, is not constrained to the use of standard questionnaires or specific types of questioning (Arshad et al.,

TABLE 3 Summary of MMAT quality appraisal.

MMAT study design category	General appraisal	Methodological strengths	Main limitations
Quantitative descriptive	High prevalence (~30%)	High technical quality in system architecture descriptions; appropriate measurements used	Lack of control groups precludes evaluating comparative pedagogical effectiveness
Mixed methods	Moderate prevalence (~10%)	Good integration of qualitative user feedback with quantitative system metrics	Overall quality is sometimes constrained by methodological weaknesses in one of the component methods
Quantitative Non-randomized	Moderate to high quality	Robust data collection on user acceptance and short-term engagement	Mostly cross-sectional in nature; lack of longitudinal data to track multi-year outcomes
Qualitative	High quality	Strong integration of data and clear consideration of the research context (MMAT criteria 1.2–1.3)	Findings often rely on subjective experiences; researcher reflexivity is sometimes underreported

2024; Nair et al., 2022). In turn, the main benefit of such a system is its scalable and accessible nature, which provides users with a quick and convenient source of personalized career advice that caters to their skills and interests, as opposed to logistical and financial inaccessibility of in-person counselling (Lent, 2025; Monika and Balaji, 2025). Additionally, the employment of chatbots can more efficiently address the issue of anxiety and decision-making uncertainty about the future (Arshad et al., 2024; Gedrimiene et al., 2023). In an environment of a real-time and data-driven automated response that is non-judgmental by default, access at any time, the users can find comfort in self-reflection and diversification of their ideas about possible career paths (Dascalu et al., 2024; Gedrimiene et al., 2023). This is in line with the idea of AI as democratizing access by being low-cost and widely available to anyone (World Economic Forum, 2025). On the other hand, students similarly appreciate humans as career counsellors when they want to have deeper and more personal conversations, or when they need help with complex career decisions. For example, Terblanche and Tau's (2024) recent evaluation of students' experiences with an AI chatbot coach found that users greatly valued how easy it was to access the chatbot and how helpful it was for fostering self-reflection, but users specifically noted they missed the "human touch" and flexibility that a human coach can offer. Chang et al. (2024) also found that although AI mentors can quickly provide learners with the most recent career information, human qualities like empathy and supervision are irreplaceable. This implies chatbots have better use cases as supplements rather than replacements, a finding consistent with others. Westman et al. (2021) similarly found that trust and empathy were seen as barriers: students and counsellors worried a chatbot would not understand individual context or provide "genuinely personalized" empathy and care, which could limit trust in its guidance. Some strategies to improve perceived credibility, such as making it clear the chatbot is an AI agent and citing sources for its suggestions, were recommended (Westman et al., 2021). Privacy is another key consideration, since conversations with the chatbot may include sensitive personal information; users need trust that the data is kept confidential and securely handled.

In some research, however, it is noted that using AI in this field appears to be useful. Akshansh and Bhavani (2019) suggest that chatbots allow students to answer or resolve any of their queries "without the need of personally going to a career counsellor." This then would be an environment where no additional barriers are required, able to "reduce the stress of the students" with respect to their future. However, once a student wishes to make specific career decisions, or is at a level that is more detailed and personal, the paper notes that human interaction has benefits. It is seen to sustain the interest of students while using chatbots for extended timeframes would cause engagement to drop, after the initial excitement of newness is over. This sentiment aligns with findings of Pandya and Wang (2024). The AI could support the career process – helping with decision making, and answering with "valid results" questions, but not replace the human part of it. A balanced approach seems the key, and a "blended counselling" approach is how it has been described. Chatbots can be used for some guidance and triage, with a human counsellor providing a higher level of mentoring and support. This insight from a cluster of studies on AI chatbots maps on to our emergent theme of human-AI collaboration. The finding that hybrid models with AI augmenting, rather than replacing, the counsellor are most successful and trusted aligns closely. As with Driesel-Lange et al. (2025), digital tools (including AI agents) can bring value when used as part of a considered pedagogical framework. Our review corroborates and extends this argument, finding that chatbots can engage students and provide access to guidance but should do so under supervision or in tandem with professional counsellors, to ensure sensitive or more nuanced concerns are not missed.

A particularly interesting paper in this context was from Aspire Guide, published in an Atlantis Press conference paper (Bhilare et al., 2025). Aspire Guide is a web-based career guidance system based on the integration of a meta-modeling approach with a conversational AI interface. To improve prediction accuracy, the proposed system used an ensemble of machine learning algorithms such as Random Forest, Support Vector Machines (SVM), Decision Trees, and XGBoost with the best performing base models chosen through weighted voting. Results showed that the overall accuracy achieved by the developed model was 94.12%, which was significantly better than that of the individual base models. However, it is crucial to distinguish between classification accuracy and guidance utility. High accuracy on historical data may simply reinforce existing trends without encouraging exploration. The platform also included a chatbot implemented using the RASA framework which used NLP for intent recognition and entity

extraction to answer student queries with real-time personalized career advice.

From the perspective of Career Construction Theory, generative agents support the "career concern" phase by providing instant information, but risk narrative truncation if students passively accept AI suggestions instead of actively constructing their own vocational identity.

4.2 Theme 2: predictive analytics and machine learning approaches

The other main thematic area involved attempts to apply machine learning (ML) and predictive analytics for career counselling, generally with an eye toward extracting value from large amounts of data to give evidence-based career advice. Some of the articles either develop or evaluate AI models to process and analyse large-scale datasets and produce a prediction or classification. The underlying concept here is that patterns in students' data (grades, skills, interests, personality, etc.) and the labor market (required skills, job openings, etc.) can be mined for making personalized career recommendations or risk assessments. In the "big data" age, career services now have a wealth of rich data (student academic transcripts, resume databases, online job postings, etc.), and ML can be used to organize that complexity.

Some studies constructed classification or regression models that, given student information, can predict the probability of certain outcomes (e.g., getting a job in a major), or classify students into certain career categories. One example is the framework by Guleria and Sood (2023) to support career counselling using a framework that they call educational data mining. They trained several classifiers on students' academic and skill attributes to predict a job sector or the probability of placement for each student. Their focus was also on Explainable AI (XAI), i.e., not only high performance but also explainability of the model's predictions. In their evaluation, they found the Naïve Bayes classifier to perform the best (with a Recall of ~91% in predicting whether a student got a job in their desired field). However more importantly, the XAI framework allowed the system to explain its predictions (e.g., to show which skills or grades were key factors in the recommendation), which is important in a counselling context for user trust. This reflects a more general point: while high accuracy is important, in career guidance, a "black-box" prediction without explanation is of limited value, a student and counsellor should be able to understand why a system makes a certain career path recommendation (Pandya and Wang, 2024). For this reason, many of these studies emphasize model interpretability. Some of the specific techniques included: decision trees (which by nature are more interpretable), rule-based expert systems with some ML augmentation, and visual analytics for users to see data patterns.

Empirical research has subsequently exploited EDM techniques for predicting graduate employability. Previous research has shown that the accuracy of using prediction techniques such as Decision Trees and Bayesian networks (machine learning algorithms) in predicting employment status was higher when compared to classical statistical algorithms. Mishra et al. (2016) and Othman et al. (2018) have developed models that augment features to include not just performance (GPA) data, but also other behavior-based data such as participation in extra-curricular activities and completion of internships.

The incorporation of predictive analytics allows the possibility for scalable intervention models. For example, Georgia State University (GSU) is able to determine at-risk students through data and offer early, personalized career coaching in an effort to support and guide students early in their career development (Shanks, 2024). The data-driven approach offers an easily scalable model across different institutions since the core technology and analytics can be modified for different student bodies and institutional setups. Moreover, the most effective career services practices are partnering with other institutions and employers. One example of this is Morgan State University's "Tomorrow's Innovators" program, a collaboration with Bristol Myers Squibb that supports a talent pipeline for underrepresented students in the biopharma industry (Shanks, 2024). In these cases, predictive technology is embedded within university services, and when paired with partnerships across industries, offers scalable preparation for students in high-demand jobs.

Moving beyond prediction to prescriptive analytics, more recent work has begun to develop systems to facilitate job recommendations. A leading example is the collaborative filtering model by Yao et al. (2023), which addresses the "cold-start" problem by matching students to alumni with demonstrated success based on skill similarity. Al-Dossari et al. (2020) further proposed CareerRec, a targeted system trained on the profiles of thousands of IT alumni. Their results showed that the XGBoost algorithm attained ~70% accuracy for recommending role types to seniors (e.g., developer, systems analyst), which their work interpreted as evidence that alumni serve as accurate predictors for current students.

An example of predictive system is the one by José-García et al. (2022), who proposed C3-IoC, an AI-based career guidance tool for IT students. It uses a new similarity-based ML technique to map a student's skill profile onto job roles in the IT domain. It then visualizes a network graph of different job roles with the student's own competencies, showing how similar or different the student's skills are from those in each job role and highlighting the skill gaps for jobs of interest. This predictive approach thus does not just recommend jobs but also which skills students should improve in to better fit a job. In their preliminary evaluation, students found the visual feedback useful for self-assessment and planning (effectively an AI-augmented self-assessment tool). Similarly, a recent study by Majjate et al. (2023) introduced a system for academic guidance that also uses a predictive model to not only recommend careers, but also academic pathways (which elective courses, specializations to take in order to reach a career goal), claiming high R-squared value in model fit. These illustrate how predictive analytics can be extended from career counselling into academic advising, closing the gap between what students' study and where they can work.

In sum, the predictive analytics theme illustrated how AI could help make data-driven predictions to inform students' and counsellors' intuition with career decisions. By examining patterns too large and complex (particularly with multidimensional data sets) for humans to perceive, such tools could help discern non-obvious matches. Especially relevant for the ethical considerations' discussion, it is also key to note how the adoption of XAI was a key methodological innovation in the predictive analytics space, as it is well-aligned with the view that counselees should have transparency and remain the final arbiter in career decisions. In the discussion section, we will further connect our findings on predictive systems to those noted by Pandya and Wang in their scoping review of AI/ML in career guidance, particularly in relation to the call for a more critical adoption of AI with attention to potential risks (Pandya and Wang, 2024). In fact, our review suggests that researchers on predictive systems are already

quite aware of those risks and are developing protective safeguards (such as interpretability) into their models.

4.3 Theme 3: recommender systems and personalized pathways

The third theme included recommender systems and other personalized learning and career guidance AI platforms. This category has some overlap with the predictive analytics approach but is distinct in that these systems are often the same thing as a full-fledged digital career guidance platform, an AI-enabled comprehensive, integrated service platform, for all things career exploration.

The best example in our sample was Sliusarchyk (2025), who detailed a digital career guidance system for IT professions. The system comprised several modules: an adaptive user questionnaire that learns on the way and which students fill out to provide a complete picture of their interests and competencies, an analytical core which houses algorithms (classification of IT professions, rules engine) that process and analyse this data to arrive at recommendations, a knowledge base that holds information about career paths, professions, and the competencies required for each, and a user-facing visual interface to support career navigation. The system was intended to overcome some of the limitations of international platforms such as O*NET or ESCO: it focused on more fine-grained IT specializations and mapped in educational pathways. The platform was able to be linked to Learning Management Systems (Moodle or Canvas, for example) to factor in a student's actual academic history when producing recommendations. Sliusarchyk (2025) evaluated her system in context (such as in a university career center) and even in a proof of concept in a municipal youth agency. The paper did not have full empirical evaluation results at that time, but it is a good design example of how AI recommender systems can offer a systematized and modular approach to career guidance, an ecosystem in the sense of bundling together interest assessment, labor market info, educational advising, and so on.

A more technically involved approach in this vein was the use of ontologies and multi-agent systems to build a quasi-intelligent ecosystem for career guidance (El Haji and Azmani, 2020). This work described a system design based on multi-expert, multi-agent architecture: there are different AI agents that each specialize in one type of expert function (one agent suggests occupations, one suggests relevant training programs, one agent monitors labor market trends, and so on), and the agents coordinate with one another to jointly arrive at guidance recommendations. This is undergirded by an ontology of career-related concepts so the system "knows" semantically how these concepts interrelate. For example, an ontology may encode that "Data Scientist" is a type of occupation that requires the skills of programming, statistics, and domain expertise in an area (finance, healthcare, etc.). When a student provides input to the system, the agents interrogate the ontology and the big data sources to generate guidance. El Haji and Azmani (2020) emphasized that such an ecosystem is built to consider all the actors (students, parents, teachers, counsellors, labor market stakeholders) in the guidance process, and this was their way of incorporating a governance perspective. This holistic approach would (according to these authors) lead to stronger guidance, because it grounds recommendations by triangulating the inputs of multiple actors and is modular and scalable. This paper is mostly conceptual but with a prototype and is interesting because it explicitly took up the question of platformisation, the coordinating role of the platform in managing the guidance-related activities among stakeholders. This topic was recently analyzed from a policy perspective by Staunton (2025) in the context of platformisation in career development more broadly. El Haji and Azmani's (2020) ecosystem can be understood as an attempt to design a benevolent platform that orchestrates career guidance, rather than letting students fend for themselves in the wild with disparate tools (job boards, university websites, etc.) and information.

To overcome the limits of static guidance and to address, at a minimum, the high ratio of students to counsellors that Dutta et al. (2022) have pinpointed, at university level, career counselling recommender systems have been shifting toward hybrid architectures that connect collaborative filtering with content-based and rule-based engines of inference. As Mavuso et al. (2023) discuss in terms of personalized guidance, the positive aspects of AI-founded customization in education outweigh potential harm. Current platforms rely on taxonomies that intersect student profiles (backgrounds, skills, and unexpressed interests) with labor market data in mappings between learners and career pathways. Jiang (2025), for example, has designed a dynamic system informed by Markov decision processes, which first delimits a given user's "state space" and then, in the face of changes to either environment or skill development, reshapes the recommendations in real time. Birzaneanu et al. (2024), in a paper on "CareProfSys," discuss a platform that, through ontological inferences and the HermiT engine, segments the student population according to the data gathered from social media profiles, CVs, and personality tests but also uniquely "fine-tunes" its recommendations by offering mini-games in Virtual Reality (VR). This hybrid logic not only allows for data-driven recommendations, but it also offers pedagogical coherence: an immersive, iterative experience of the user's interaction with real-world outcomes of jobs, therefore supporting students' agency and engagement (Birzaneanu et al., 2024).

Studies demonstrate that AI-based career counselling systems could improve students' career decision-making self-efficacy to an extent similar to traditional face-to-face methods. Experimental evidence also shows that these systems help to expand students' career exploration in terms of more exploration of options and less indecisiveness (Leung, 2022). In addition, AI-guided interventions, especially when implemented through a motivational lens, have achieved strong user engagement and satisfaction in the university context (Han et al., 2025). Another evaluation, from Spain, of a system recommending vocational training opportunities based on skills held found that more than 70% of users found the recommendations relevant and helpful in discovering new options (José-García et al., 2022). However, user responses to such systems have not all been glowing, some users of the systems in these studies felt that recommendations could be too generic, or not take into account personal passion. So, for example an algorithm might obviously guide a high-achieving student in chemistry toward careers like lab researcher or chemical engineer, but neglect less obvious and potentially more appealing paths unless the student had somehow articulated that in their profile. This is a reminder of the challenge of capturing the whole person in data form; a challenge we will take up again in terms of designing for the human agency of the AI-student interaction.

This theme also brought up ideas of comprehensive career development ecosystems for supporting lifelong guidance. Driesel-Lange et al. (2025) highlight the need for new and inclusive digital tools and report that collaboration between educators, researchers, and digital experts is an important part of that process. The AI-based platforms

fit this vision in that they are spaces where various stakeholders can plug in. For example, career counsellors can input their expert rules, employers may input labor market information, students get personalized outputs. Some platforms allow career counsellors a dashboard view of what students are receiving as suggestions or to tweak the system rules. The counsellor, in effect, can "fine-tune" the AI, which is a promising practice for blending in expertise. In the latest study, Kulugh et al. (2025) designed an AI-based career guidance system that combines Holland's Vocational Theory and Social Cognitive Theory. Large Language Model APIs (Llama) were used to mine the user's profile and generate relevant recommendations for the user, with the ability to analyse "massive" amounts of information which was not possible with standard methods. These findings support the Platformisation hypothesis (Staunton, 2025): by filtering options based on opaque algorithms, recommender ecosystems act as architectural gatekeepers, potentially narrowing the student's horizon to historically popular paths.

5 Discussion

The integration of AI into university career services heralds more than an incremental improvement in delivery, it marks a paradigmatic shift in how the guidance relationship is configured. In addressing our first research question regarding the types of implemented systems, our review identified three distinct typologies predictive analytics, recommender ecosystems, and generative chatbots. Each of these reconfigures the counsellor-student dynamic differently. To understand these shifts, we conceptualize our results in the theoretical space defined by Career Construction Theory (Savickas, 2013) and Platformisation (Staunton, 2025). We posit a latent tension between how platform architectures may upend the scalability of algorithmic interventions and how such interventions mediate student agency.

On one hand, from the perspective of Career Construction Theory, the clear theoretical advantage of predictive AI is that it may help enhance adaptability. By mapping students' competencies to emergent job clusters, the digital scaffolding effect of such systems may help students better traverse the high volatility of the 21st century labor market. Indeed, the reviewed studies point to such systems as useful signals that externalize the information-dense task of data synthesis for students. However, addressing the pedagogical outcomes (RQ2), the results reveal a critical tension. In the bargain for such efficiency, we also see the same studies identifying the risk of automation bias, with students ceding to high-performing models an "epistemic authority" over their own instincts. If AI guidance tips over into the over-supply of definitive signals that students trust without question, the very meaning-making that constitutes the pedagogical core of the student-development paradigm risks being short-circuited. When the career-development journey becomes a matter of querying an algorithm for "yes" or "no" answers to an identity question, then the model, trained solely on easily quantified and sourced metrics like "skills," takes the singular role of teacher, assimilating the student's narrative into a data processing event. Westman et al. (2021) note this issue explicitly when they state that when a student interacts with an algorithm that gives a career path based solely on the "measurable" aspects of their academic record, the multidimensional narrative of who the student is as a person is flattened into a series of points on a vector space. Thus, a key consideration for educators designing AI-based career interventions is how to implement these systems so that AI augments the decision-making process but does not take over the decision-making agency.

The results in this study further support Staunton's (2025) framework of platformisation by extension: by governing the mediation of the guidance relationship, career guidance is subject to an increasing "hidden curriculum" of platform architecture. Platformisation of guidance leads to an inevitable transfer of epistemic authority to the gatekeeping mechanisms of AI models. Recommender systems trained on certain data to favor high-demand economic roles, or ones that use training data that simply reflect legacy institutions' historical training programs, may bias students away from non-traditional trajectories. The high degrees of personalization touted by user profiling approaches are in this way a kind of algorithmic determinism, constraining the breadth of the student's exploration with a calculated façade of student voice. The recent scoping review mentioned earlier in this study (Manganello et al., 2025) echoes this trend, singling out fairness and transparency as particularly salient blind spots in extant educational AI literature. When platforms do not offer Explainable AI that students can interrogate, the recommendations for pathways become recommendations in all but name, as students cede decision-making to a seemingly omniscient source. It therefore becomes incumbent on the institutions to govern these platforms: that is, data ownership and other data rights, their choice of (transparent) algorithms, and how they train and retrain models to reduce bias (Staunton, 2025). These elements of platform governance are as much a part of their pedagogical approach as a syllabus or teaching philosophy. The contextual significance of these findings is that institutions must enact robust platform governance. Demanding XAI and ensuring data portability are no longer just technical requirements, but essential pedagogical responsibilities to protect student agency. This interpretation aligns with and extends prior studies that highlight fairness and transparency as critical blind spots in AI-driven support systems (Manganello et al., 2025; Pandya and Wang, 2024), demonstrating empirically that without transparent governance, AI tools risk perpetuating historical biases rather than expanding vocational horizons.

Generative AI adds a new wrinkle to this paradigm by migrating the conversation from a search mechanism to a chatbot. It is not difficult to see from the evidence presented here that student use of generative agents is likely a function of perceived utility and the student's current information need (Wang and Li, 2024). Theoretically, a well-architected generative agent provides a 1: 1 scaffolding function, and a scalable, conversational mode for students to co-construct their vocational narrative in real time. However, as mentioned earlier, there is an attendant "halo effect" in generative AI's fluent, anthropomorphized form of interaction that could lend undue gravitas to agents' synthetic narratives. In contrast to a human counsellor's ability to ask questions around a student's emotional response, or to probe for eccentricities that signal core values, a generative model has little way to look outside pre-selected response sets or the normalizing tendencies of its training distribution. A clear danger, therefore, is that a student may take on and internalize a synthetic narrative as a self-narrative. As a solution, human-empowered approaches identified in this paper tend to position generative agents as a chatbot-first, human-second line of support: for example, employing them for handling FAQs, following up, or role-playing, but not for the psychosocial work of helping students self-define or construct a narrative (Wang and Li, 2024).

Taken together, AI's role in university career counselling must move beyond a "technological solutionism" to orchestration of the pedagogical relationship. Crucially, our analysis of the evidence base (RQ3) highlights that most current implementations are prototypes lacking longitudinal validation. Consistent with this paper's results, we find that AI tools are best employed as "boundary objects" that students and counsellors use to augment conversation (Guleria and Sood, 2022). As such, future design must be careful to empower students to critically question recommendations rather than accept them as truth, for which institutions must provide Explainable AI to build AI literacy (Guleria and Sood, 2022). In the same way, they must also provide institutional backstops like privacy-preserving but cross-institutionally portable data ecosystems to ensure that AI personalization never sacrifices student agency. In the end, while AI has much to offer the scale and customization of university career services, its implementation must be done with an abiding recognition of human agency if the technology is to serve as anything more than a tool in the student's lifelong endeavor of career construction.

6 Future directions

According to the results of this review and the particular methodological constraints raised in our assessment of quality, we propose some directions for further research:

(a) Longitudinal and quasi-experimental studies: Longitudinal follow-up and randomized controlled trials need to be conducted to assess whether AI interventions do indeed enhance graduate employment for long-term, and sustained vocational adaptability.
(b) Evaluating XAI and AI Literacy: As automation bias presents a tremendous threat to student agency, the way research on XAI interfaces impact a student's critical thinking and contribute to building AI literacy must be empirically evaluated.
(c) Platform governance and algorithmic audits. Researchers should conduct critical algorithmic audits of recommender ecosystems to understand how algorithmic "fit" is operationalized, so that algorithms do not disproportionately guide marginalized populations along historically biased paths.
(d) Human-in-the-Loop Frameworks. Pedagogical research needs to test hybrid counselling models in this way, comparing fully autonomous AI guidance to AI-augmented human counselling and figuring out how best to calibrate technology efficiency with human empathy.

7 Limitations

Before we finish, we must acknowledge the limitations of our own review. First, despite our best efforts at comprehensiveness (notably an exhaustive search), we may have missed relevant studies. The focus on studies up to the year 2025 means, of course, that the field will have moved on from what we have captured (AI is indeed a moving target). Second, the heterogeneity of studies and the qualitative nature of much of the evidence precluded a formal meta-analysis. We had to synthesize the findings narratively. We did this with rigor through the use of MMAT and careful extraction, but there is always an element of subjectivity in how themes are categorized and implications of studies are interpreted. Third, we included only implementations for university students. We did not address the wider context of AI tools that students could (or might) use unconnected with their studies (e.g., LinkedIn's use of AI features, general apps), except where they are part of research studies. Students' use of AI tools is a related but separate area that could also benefit from review, but was outside the scope of this one. Additionally, because this review draws on diverse literature, including both empirical evidence and conceptual discussion (e.g., platformisation) we have occasionally extended the discussion beyond what the included studies directly address to infer what they tell us about that. We have strived to ensure that such discussion is firmly rooted in a cited source or a logical extrapolation, but readers should be aware of that blend of evidence and interpretation.

Another limitation is the focus on English-language studies only. Although English dominates educational technology research, limiting searches to this language biases against AI implementations that have been documented in domestic languages only. As such, our review may be biased toward a "Western-centric" or Anglophone understanding of how AI gets adopted, potentially missing highly localized models of algorithmic career guidance.

8 Conclusion

The adoption of AI in career counselling for university students has transitioned from a series of pilot projects to a widespread practice within career services. Spanning 2015–2025, this systematic review shows that AI-powered tools such as chatbots, predictive analytics, and recommender platforms can meaningfully enhance career guidance at scale through "scalable personalization." Through automation, career centers with limited resources have provided personalized advice to many students at once, improving engagement and offering data-driven guidance.

At the same time, this review underscores that AI does not function as a complete "solution" or a substitute for human counsellors. Rather, the emerging practice is one of complementarity. AI excels at data processing and pattern recognition but requires human counsellors for empathy and complex emotional dynamics. The most successful initiatives so far have been hybrid approaches that combine the efficiency of AI tools with human oversight. This conclusion is consistent with Pandya and Wang's (2024) recent review. It goes a step further by providing empirical support that AI should take on more routine work to free up counsellors to provide deeper, more meaningful support to students.

Compared to the recent work of Pandya and Wang (2024), Driesel-Lange et al. (2025), and Herath et al. (2024), this systematic review reinforces many of the same implications of AI, including both the opportunity and potential dangers of AI. It provides a needed update to previous reviews, for example showing the recent trajectory of AI-based tools toward greater personalization. It also contributes important empirical specificity to understand how concerns about ethics are playing out in practice. While around 43 studies may not sound like many, that is quite a strong knowledge base in a newly emerging field to provide for evidence-based practice.

In conclusion, this systematic review functions as a helpful guide for the responsible adoption of AI solutions. Such adoption is possible

but requires deliberate design and human oversight to prevent dehumanizing and potentially biased outcomes. AI is not the culmination of the "digital transformation" of career support, only one step along an ongoing evolution. Responsibly deployed, AI has the potential to provide students with powerful tools to meaningfully and effectively navigate their educational and career pathways, as long as student wellbeing remains a central concern in innovation efforts.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

YS: Conceptualization, Resources, Investigation, Writing – review & editing, Writing – original draft, Methodology, Supervision, Formal analysis. AA: Data curation, Writing – original draft, Conceptualization, Methodology, Writing – review & editing, Supervision. AK: Project administration, Data curation, Writing – review & editing, Investigation, Funding acquisition. BU: Data curation, Investigation, Resources, Writing – original draft, Formal analysis. IS: Validation, Methodology, Writing – original draft, Investigation, Resources. AB: Data curation, Formal analysis, Investigation, Writing – original draft.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was not used in the creation of this manuscript.

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Supplementary material

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